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A Future War in SF and Posthumanism

SF 속 미래전과 포스트휴머니즘:
Terminator 2와 The Matrix를 중심으로

이 한 솔(Hansol Lee)
육군사관학교 영어과

ABSTRACT

Technology has dramatically changed the nature of warfare. In particular, the exponential development of technology has spawned the creation of Artificial Intelligence (A.I.) not only for the benefit of humans, but also for the development of military weapons system. Substituting a fighter-wing with a drone, modern technology for future warfare may help avoid some of the collateral damage in a battlefield. This innovation, however, has brought about both conveniences and inconveniences: the intelligence of "smart" machines is, to some extent, equal to that of humans. This development can lead to conflict between Human and Machine, a possibility depicted in two Science Fiction movies: Terminator 2: Judgment Day and The Matrix. Humans are still regarded as the only rational, autonomous, unique, and freely existing beings. However, what happens if A.I. can analyze the complicated emotions of humans and simulate these signals in a computer program? Unless the conflict between a computer and its user is resolved, a war between Human and Machine will emerge sooner or later. In accordance with these questions and problems, this paper analyzes the two movies mentioned above to focus on the inevitable future warfare or conflict between Human and Machine. Moreover, it further delves into Thomas Pynchon's novels including The Crying of Lot 49 in order to graft a possible solution for future war.

Key words: Future War, Terminator 2, The Matrix, Thomas Pynchon, The Crying of Lot 49, Posthumanism, Artificial Intelligence (A.I.), Science Fiction (SF)

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Since human beings began to stand on two feet and their two hands became free to move, various technologies have been invented to control and adapt to the natural environment. The Machine Age, considered to be at peak between the two World Wars, emphasized the vitality of the machine. Corollary to this development, human beings have gained numerous benefits and conveniences. Even though the "technology is essentially a form of service" (Lanier 1), the distance between Human and Machine has become so close that the two can no longer be separated (Leo Marx and Merritt Roe 4). Yet, this is part of the stream of our historical movement. As Katherine Hayles explains, "every food that humans currently eat has already been the product of such technological interventions [of Machine] as selective breeding" ("Afterword: The Human in the Posthuman" 134).

Indeed, the advent of Artificial Intelligence (A.I.) has brought about a huge development in civilization. However, while A.I.'s become increasingly smarter, the distinction between Human and Machine has also become more tenuous. Now, it is undeniable that highly advanced computers can analyze even the complicated emotions of human and even simulate these signals within computer programs (Hayles, How We Became Posthuman 2-3). In other words, Machine can think. In the unstoppable development of computer technology, the conflicts between two "species" would continue to escalate unless mighty God somehow reverses the course of history.

On the other hand, some philosophers, including Neil Badminton, inform us of the ominous assertion that the radical break of humanism can lead to "apocalyptic posthumanism" (quoted in Simon 5). Also, Badminton regards brain death as the end of human life: he asks us, "if 'Man' is present at 'his' own funeral, how can 'he' possibly be dead?" (quoted in Simon 5). What a conflict! Furthermore, there are many doubts on current technology as well. They say it is a part of common sense that human's intelligence is regarded far from algorithm system which composes machine's processor including A.I. That is to say, technology only can represent a result calculated by algorithms of "statistics and correlations," not by "emotion or meaning" as usual humans

do (Lanier 1). According to Lanier, a degree of understanding seems to matter in distinguishing Human and Machine under the present circumstances, and he continues that this is the reason "why we always look both ways when crossing an intersection even though the light has turned green" (1).

Nothing can make all doubt disappear about development of technology toward human's uniqueness, although its exponential growth will continue. However, it seems that Lanier's theory may lose its persuasive impact when a critical point has it that humans are becoming machine-like. As Hayles states, humans are becoming data-processing systems much like computers, and "consciousness" is just one of the many properties that can "increase the functionality of the system" (How We Became Posthuman 238).

At this point, to the question what is the humanity to define Human, Michel Foucault claims that "Man is neither the oldest nor the most constant problem that has been posed for human knowledge" (386). He continues to state that "man is an invention of recent date" which is "nearing its end" (Foucault 387). In other words, Human is not an absolute truth that cannot be altered. As Foucault deconstructs the man, Donna Haraway suggests cyborgs as a model of posthuman subjectivity, saying it is "a cybernetic organism, a hybrid of machine and organism" (149) that blurs the borders between "self and other, male and female, organism and machine, ontology and textuality" (Bartlett and Byers 28).

When it comes to future warfare, the improved A.I.'s can be both our friendly weapons and unavoidable enemy. For instance, an artificial heart may prevent and cure people's cardiopathies. Also, artificial bones are becoming an alternative to replace human bones that are damaged due to fractures, disease, etc. Then, what about an artificial brain? What if it can learn and study to behave and conduct as people do? In the same sense, high-tech weapons such as drones or robots can replace conventional weapons system, but what about cyborgs with conscious thinking microprocessor as our national defense system? Probably, we need to think about it thoroughly.

Actually, in the military science area, "information warfare" is considered to be a

possible leading future warfare (Thompson 1). In other words, humans will wage wars using computers to destroy enemy's C4I system—including civilian and military infrastructure of an enemy—through attacking network system, the Machine. And it proves that man cannot survive without machine now and that the possibility of the conflict between the two turning into a war is never small.

Due to this momentum, the idea of close relationship between Human and Machine and of reciprocal influence is common enough to infiltrate into mass media (Schwab 69). For instance, the movie Terminator 2: Judgment Day (1991) suggests a possible future war between the two, where, in 2029, computers destroy and massacre human civilization. In the same vein, The Matrix (1999) illustrates a world where Humans have been degraded to become tools to produce electricity and a source of power for Machines in a computer-dominated world called the Matrix. Thomas Pynchon's The Crying of Lot 49 depicts a woman who travels to San Narciso, a town built like an electronic circuit that was affected by a computer company. She can only think four ways—00, 01, 10, and 11—under the digital system of the influential computers.

In fact, as Damien Broderick says, science fiction is "storytelling native to a culture undergoing the epistemic changes implicated in the rise and supersession of technical-industrial modes of production, distribution, consumption and disposal" (quoted in Roberts 12). So to speak, watching Terminator 2 and The Matrix and reading The Crying of Lot 49 we can be "conscious of how we are evolving as a species in the present by speculating about where we are headed" (Roe 312). For this reason, this paper aims to delve into two SF movies and one science fiction to help us understand what man is now, can be, and should be.

Ⅱ.

It might seem that both movies are just following formal distinctions between Human and Machine, but they have rendered such conceptions obsolete. These films blur the substantial boundary between Human and Machine. Especially in Terminator

2, humans such as police officers or psychiatrists, who are obliged to protect others, are depicted as if they do not have a heart which is believed to be a basal condition for "man". Rather, terminator T-800 supports and guards John Connor and his mother like a human with emotion.



Terminator 2: Judgment Day

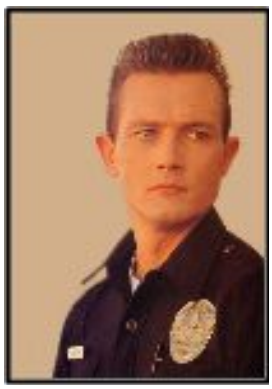
(Terminator is a cybernetic organism. Living tissue over a metal endoskeleton. In other words, it seems—this poster apparently depicts the point already—as if terminator is a human and machine at the same time.)

This is why this movie stars terminators as its heroes. And this is why terminators take human's appearance making it hard to differentiate from machines. While T-800 burns himself to death to save Human, Doctor Silverman never listens to Sarah (John Connor's mother) when she honestly confesses her authentic knowledge of upcoming events. At the same time, the police chases John and Sarah, while they help T-1000, a terminator from Machine to kill John to put Human in jeopardy, just because the boy was an official juvenile offender and his mother was a prison breaker from a mental hospital. Doctors and policemen see her as merely a psychotic patient. The doctor's medical knowledge that is believed to be one of the absolute truths, and police's credo "To Serve and Protect" rather turn them into machine-like entity which only follows its own algorithm. Therefore, the primary conflict in this film is among humans themselves. And it directly represents the severe problems of communication or reciprocal understanding in human society. When the Connors try to prevent the apocalyptic fate by A.I. in national defense system, their efforts are ironically deterred by Human. At the same time, when Machine's terminator, the T-1000 steps in and moves to kill John Connor, it is the T-800, another machine manufactured by Cyberdyne, that blocks

the T-1000's mission to terminate him.



T-800



T-1000

(Actually, Terminator 2 blurs not only the distinction between Human and Machine, but also good and evil: a tough bike riding gangster saves human while a policeman of neat figure kills them.)

Turning into a human-like-being, T-800 represents attributes of A.I. that can both save and destroy humans in compliance with our behavior. That is to say, when mutual understanding between the two is possible enough to recognize or to understand each other. As a matter of fact, T-800 was originally programmed just enough to carry out few necessary missions. With its A.I. processor, it can learn and study to enhance its capability to initiate the missions. Ironically, its microprocessor, the essence of A.I., is the only possible condition for John to educate T-800 not to kill people when it kills the innocents without any reasons or guilty consciousness. However, it turns out to be a moral entity which does not kill people anymore (terminator's originally programmed mission is to terminate objects under any circumstances without any hesitation), instead it becomes a humane creature who can wipe away John's tears, although it cannot weep. Thus, T-800 becomes a hero who can sacrifice itself for others. As a result of their mutual acknowledgement, it changes Sarah Connor's ludditish perspective and allows her to regard him as a father figure to her only son.

> Watching John with the machine, it was suddenly so clear. The terminator [T-800] would never stop, it would never leave him. . . . it would always be there. And it would not hurt him, never shout at him or get drunk and hit him, or say it couldn't spend time with him because it was too busy. And it would die to protect him. Of all the would be fathers who came and went over the years, this thing, this machine, was the only one who measured up. In an insane world, it was the sanest choice. (*Terminator 2 transcript*)

The movie depicts Sarah as a typical Luddite due to her old trauma against the Machine, following her terrifying experiences in Terminator 1. When John Connor and T-800 take a short break from eluding T-1000 which was chasing them, Sarah attempts to destroy the microchip in T-800's skull to eradicate the helps of Machine civilization. Later on, however, she recognizes him as Human. Sarah's awakening opens her closed boundaries of knowledge and allows her to rethink her concept of "man."

Watching Terminator 2, the distinction between Human and Machine collapses into a meaningless classification in the future war. In the same sense, The Matrix delineates the conflict between Human and Machine as the structure of Neo vs. the Matrix, showing its hero is Neo, not Mr. Anderson: Neo is a cyberman or "an unstable, impure mixture without discernable origins" (Simon 4). As Laura Bartlett and Thomas Byers state, "The Matrix places posthuman subjects at the center of its action" (30). Likewise, Mr. Anderson in Matrix displays his computer skills to save humans "like a machine" in a computer system using his online nickname, Neo: You can define Neo either as a pure human or machine.



The Matrix

(This poster depicts Neo's view after recognizing the reality in Matrix. He can see the computer program codes that form their world through his eyes.)

In this movie, human beings are enslaved by A.I.'s after losing a war between Human and Machine. They do not feel a thing, and have been kept asleep to provide electronic power for computers, the dominant authority in this world. The Machine, in this case the Matrix, bestows "reality" to Human through electronic illusion: a disembodied consciousness in which the sleepers cannot recognize and see the real world. As a result, human beings in this movie manage their lives within the Matrix without knowing the reality just like Doctor Silverman or other ignorant humans in Terminator 2. Therefore, there is no way to restore the communication gap between Human and Machine.

In the middle of this computer world, however, there is our hero, Neo, who seeks the answer for future war. With the aid of Morpheus and his own choices, he begins his journey to see through the "reality" because, as Morpheus claims, the Matrix "pulled the wool over your eyes to blind you from the truth" (The Matrix transcript). As Bartlett and Byers state, humans have become blinded by "the mind forged manacles of the Matrix" (37).

Perceiving reality, Neo becomes the "one" who can connect Human and Machine via network as a way of communication. Only inside of the Matrix can Neo demonstrate his ability to save the Humans. He does not, however, change or terminate the system, but only watches the reality of the Matrix and recovers severed communication between humans and the Matrix. Neo does not give us any specific answers or salvation, but urges us to realize the real "reality" and then, states that the choice is ours. At the end of the movie, Neo flies over the earth to seek the answer to question that he still does not know, and he finally communicates with the Matrix.

> I know you're out there. I can feel you now. I know that you're afraid. You're afraid of us. You're afraid of change. I don't know the future. I didn't come here to tell you how this is going to end. I came here to tell you how it's going to begin. I'm going to hang up this phone and then I'm going to show these people what you don't want them to see. I'm going to show them a world without you, a world without rules and controls, without borders or boundaries, a world where anything is possible. Where we go from there is a choice I leave to you. (The Matrix transcript)

Now, he urges people to see the reality what they cannot understand or do not want to know. In other words, providing information of Machine, Neo makes Human recognize the existence of the Matrix and the conflicts between themselves, helps connect them by becoming a stepping stone. This then will only be another beginning, not the end, because the choice is left to us. We can then understand each other and ensure the future without ignorance or misunderstanding.

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In conclusion, the two science fiction films illustrate the acceptance of the difference and reciprocal understanding between Humans and Machines. However, if there is anything that is beyond doubt, conflict between Humans and Machines is inevitable. In Terminator 2, John survives and will become a leader of the resistance against the Machines someday, after the computer system's revolt and becomes smart and confident enough to think for itself. Neo does not attack or neutralizes the Machines; thus, Humans remain the battery packs of computers in The Matrix. In other words, the movie sets this conflict as an unavoidable premise.

In the end, the movies propose how to recognize, prepare, and cope with this historical movement, but they never suggest any ideal alternatives or portray Utopia. Thus, it is timely and appropriate to delve into Pynchon's story to understand the movies' message because it mainly discusses the problem of communication between conflicting groups. The writer, who majored in Engineering Physics at Cornell University and worked at Boeing, allowed his scientific and technological insights to merge in The Crying of Lot 49. Although this novel does not deal with future warfare as do the two movies, it does point out the closer relationship between Machines and Humans in designating mechanical patterns, which is one of Machines' typical attributes, in human society and emphasizes the importance of reciprocal understanding to recognize the others. Moreover, Pynchon's story does not propose a rosy future but an endless waiting for the answer as Terminator 2 and The Matrix presuppose: conflict between Machines and Humans is unavoidable.

This story is about Oedipa Maas, who takes it upon herself to shoulder her former boyfriend's inheritance problem. During her journey, she gains new insights and ideas about peripheral circumstances. On the surface, her investigation throughout the journey appears to be a novel's main point; however, she mainly confronts her limited knowledge of Human throughout the story. Indeed, the future war the two movies depict is similar to the problem Oedipa encounters during her journey. At the very end, "man as rational, autonomous, unique, and free, is in turn challenged and deconstructed" (Simon 4). Just as in the two movies, she finally determines a solution and sees the real reality. Linking the severed communications, she finds herself in a system consisted of the dangers of uncritical and naive thought towards absolute truth. In short, the novel indicates that we need to recognize unlimited possibilities, and not to come up with hasty conclusions or make decisions without recognizing things in blind side. Without changing our cognition and going against established boundaries of knowledge, there will be no creative or helpful future, as Oedipa experiences through her journey in closed system.

In a similar manner, the future war between Human and Machine in Terminator 2 begins when Cyberdyne recognizes itself as a subject, when A.I. of Cyberdyne makes the supposition that Human is the most threatening object, which is similar to how Human regards Machine with communication gap. At the same time, in The Matrix, the traitor Cypher, who only sees illusions over reality and pursues virtual pleasure, and Agent Smith, who cannot see the physical world and rejects reality, are the trouble-makers in the Matrix, the crucial reason for the endless conflicts between the two. When mutual understanding and communication is severed and entropy begins to rapidly increase, the world will become a cul de sac. This, as two movies illustrate, is the future. As a result, "the muted postal horn" should be able to cry in a stately manner for the Tristero to be recognized. Indeed, communication is the only key (Schaub, Pynchon: The Voice of Ambiguity 77) to approach high-tech future warfare.

Basically, The Crying of Lot 49 does follow the idea of the conflict between Machine and Human. In particular, it also points out the closer relationship between the two parties designating mechanical patterns, Machine's attribute, in human society.

She [Oedipa Mass] thought of the time she'd opened a transistor radio to replace a

> battery and seen her first printed circuit. The ordered swirl of houses and streets, from the high angle, sprang at her now with the same unexpected, astonishing clarity as the circuit card had. Though she knew even less about radios than about Southern Californians, there were to both outward patterns a hieroglyphic sense of concealed meaning, of an intent to communicate. (14, italics mine)

Moreover, she feels as if she is "walking among matrices of a great digital computer, the zeroes and ones twinned above, hanging like balanced mobiles right and left, ahead, thick, maybe endless" (150) because of the influence of Machine. When Oedipa is wandering through the endless patterns of computer system, she experiences Machine's effects on Human.

Pynchon asserts that the A.I. consists of a computer program that handles only two letters, zero and one, and it denies any possible things between zero and one. The mechanical pattern, is sufficiently strong to penetrate not only into the city, but also into the minds of humans. This is exemplified by Oedipa's search for the Tristero. She thinks of no more than four possible assumptions, the "symmetrical four" (148). If we roughly calculate the combinations of two numbers (i.e., one and zero), there are only four possible answers: 00, 01, 10, and 11.

However, Pynchon seeks a hope under this conflict between the two. Maybe this novel only follows the passive attitude that was shown in the two movies. It seems that Terminator 2 screens a tenuous movement to save John at the unstoppable and untouchable predestined clash between the two, while The Matrix depicts the situation that Neo could not cease the war, but just maintains the status quo. In other words, both pictures would not topple a world, but seek to achieve a defective equilibrium.

Not only does Pynchon reveal the negative effects of Machine, he also acknowledges its positive side. In his essay, "Is it O.K. to Be a Luddite?" he narrates that "with the proper deployment of budget and computer time, we will cure cancer, save ourselves from nuclear extinction, grow food for everybody, detoxify the results of industrial greed gone berserk—realize all the wistful pipe dreams of our days" (41). At first, Pynchon depicts the San Narciso, which exemplifies the negative effects of Machine, through this characteristics of Machine, represented by a computer factory, the Yoyodyne. A "yoyo" is a toy used by placing a string around your finger to make

the yoyo go up and down repeatedly. Similar to a computer system that functions endlessly with zeros and ones, "yoyo" symbolizes the meaningless repetition, or monotonous patterned daily life of humans. In Pynchon's other novel, V, "Yoyo" is defined not only as a meaningless and endless pattern of system, but also as a catalyst for Profane's restless activity.

In the epilogue of V, Profane meets Stencil's vitality just as two lines of letter V reconcile at its vertex. The repetitive nature of the yoyo heightened Profane's simple life full of ceaseless movement into meaningful and fruitful one. In addition, scientific technology has made zeros and ones produce unlimited possibilities: Cyberspace, an open space for endless possibilities. What Pynchon imparts in The Crying of Lot 49 is never to overlook anything in our world and to see through the essence of things. He does not suggest an answer, but rather emphasizes the ambiguity of things: the importance of possible other answers and the significance of communication (Schaub, "Open Letter in Response to Edward Mendelson's "The Sacred, the Profane, and The Crying of Lot 49" 101). In other words, you should not define a thing without considering all the facts that are possible because, just as Oedipa realizes, the probable answers between zero and one are hardly recognizable or illuminated.

> Decorating each alienation, each species of withdrawal, as cufflink, decal, aimless doodling, there was somehow always the post horn. [Oedipa] grew so to expect it that perhaps she did not see it quite as often as she later was to remember seeing it. A couple-three times would really have been enough. Or too much. (100)

The main reason for Oedipa's overlooking of isolated beings is the distinction set by our pre-established concepts originating from the communication gap. The dangers of seeing just one side of the fact or not seeing the isolated, ignored, or different facets is not limited to Oedipa's vulgar mistake, but also relates to the future warfare occuring in Terminator 2 and The Matrix.

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In The Crying of Lot 49, Pynchon is attempting to discover an appropriate system that could be achieved by changing our thoughts, and extending recognition to accept the diversities in a posthuman society. He never makes any promise to the reader, but rather merely suggests the possibility of an epiphany by having Oedipa sit in the auction room, simply waiting. This answer of Pynchon also lies in the title of novel, The Crying of Lot 49. Number 49 refers to the Pentecost in Christian doctrine. Pentecost is a day of celebration after the 49 days wait after the resurrection of Jesus Christ. It also refers to waiting for the Descent of the Holy Spirit, which occurs on the 50th day after the resurrection. Furthermore, in Buddhism, the number 49 also has a significant meaning in the Buddhist service, the ritual for the soul of dead on 49th day after the death. Taken together, it can be assumed that the number 49 refers to waiting for an answer that is, at the moment, being withheld. Indeed, similar to the Pentecost or the Buddhist service on 49th day, we should join Oedipa in waiting for the epiphany, while searching for the unknown results.

However, I believe the idea of waiting is somewhat insufficient to solve today's posthuman problem. Still, mutual understanding and acknowledgement is significant for possible conflicts, but Oedipa's passive waiting may cause Nihilism. Pynchon's values emphasize the complicated reality that we encounter, and the importance of considering all possible situations and facts including the blind sides, Furthermore, the waiting in auction room for the answer aims at the appropriate attitude towards the answer. An unavoidable fact, however, that Oedipa cannot find the answer remains the difficulty of obtaining a satisfactory result from posthumanism between Human and Machine. So to speak, we have to be proactive in accepting the difference and possibility as Neo chooses red pill to recognize the reality and as John and T-800 try to understand and help each other. In conclusion, we need not only to participate Oedipa's waiting for an answer, but also to encourage her to find it, armed with a better understanding, towards real reality.

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